

EU HORIZON EUROPE - APPLICATION SUMMARY

Grant reference: 101187724

Title: Climate Change, Accelerated Erosion, and the Archaeology of the Submerged North Sea Heritage Zone

Lead institution: Nordmeer Maritime University

PI: Prof. Dr. Gloria Gross

Co-PI: Prof. James Whitfield, Caledonia Maritime University

Submitted: 13.02.2026 | Decision expected: Q4 2026

1. Project Summary

The proposed project will establish the first systematic, multi-site investigation of the relationship between climate-driven oceanographic change and the accelerating degradation of organic archaeological material in North Sea wreck sites. Over a five-year programme, the German Bight component (Nordmeer Maritime University) and the Scottish waters component (Caledonia Maritime University) will deploy a paired archaeological-oceanographic methodology, combining repeat multibeam survey, sediment coring and geochemical analysis, and targeted archaeological documentation of high-risk organic deposits.

The project directly addresses a documented gap in European cultural heritage protection: the absence of quantified risk models linking oceanographic forcing functions to archaeological site preservation states. Without such models, heritage managers cannot prioritise documentation effort or justify intervention funding for the most threatened sites.

2. Work Packages

WP	Title	Lead	Months
WP1	Methodology development and baseline survey	Nordmeer MU	1-18
WP2	Site documentation: German Bight (8 sites)	Nordmeer MU	6-42
WP3	Site documentation: Scottish waters (6 sites)	Caledonia MU	6-42
WP4	Sediment coring and geochemical analysis	Both (joint)	12-48
WP5	Oceanographic monitoring deployment and data analysis	Both (joint)	6-54
WP6	Risk model development and heritage management tool	Nordmeer MU	36-60
WP7	Dissemination, stakeholder engagement, and policy impact	Both (joint)	1-60

3. Budget

Category	Nordmeer MU (EUR)	Caledonia MU (EUR)	Total (EUR)
Personnel	1240000	980000	2220000
Field operations and equipment	480000	360000	840000
Analytical and laboratory costs	180000	145000	325000

Computing and data infrastructure	95000	75000	170000
Dissemination and stakeholder engagement	85000	70000	155000
Indirect costs	416000	326000	742000
Total requested	2496000	1956000	4452000

4. Team

Prof. Dr. Gloria Gross (PI, Nordmeer Maritime University): Lead for the German Bight component, responsible for overall project coordination, WP1, WP2, and co-lead for WP4-5. Twenty years of fieldwork experience in the North Sea and published authority on Hanseatic maritime archaeology. PI of the DFG-funded North Frisian Survey since 2023.

Prof. James Whitfield (Co-PI, Caledonia Maritime University): Lead for the Scottish waters component, responsible for WP3 and co-lead for WP4-5. Published authority on Atlantic and North Sea comparative maritime archaeology, long-standing international collaborator on the North Frisian Survey.

5. Expected Impact

The project will deliver: (1) A quantified risk model for organic heritage site degradation under projected climate scenarios, expressed in terms that heritage managers and funders can use for prioritisation decisions. (2) Detailed documentation of 14 high-risk wreck sites across the North Sea, creating a permanent record of material that may be partially or wholly lost before the end of the century. (3) A methodology that can be adopted by heritage agencies across North Sea and Baltic European waters for their own site monitoring and prioritisation programmes.

The project addresses a pressing policy gap. The EU Cultural Heritage Action Plan and the North Sea Regional Convention both commit member states to the protection of underwater cultural heritage, but no risk-assessment tool currently exists for prioritising this work under climate threat. The project output will fill this gap directly.

Status: Application submitted 13.02.2026. Decision expected by 31.12.2026. Two US-based specialist contractors (remote sensing analyst; 3D reconstruction illustrator) are contracted on monthly agreements pending decision.